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Begin CommuterTransitSystem
    // Variable Declarations
    Declare String routeID, peakHour, boardMore
    Declare Integer distanceZone, passengerCount, maxCapacity
    Declare Real baseFare, finalFare
    // Initial State Setup
    boardMore = "YES"
    Output "=== FREETOWN COMMUTER TRANSIT SYSTEM ==="
    Output "Enter Route ID:"
    Input routeID
    Output "Enter Vehicle Maximum Capacity:"
    Input maxCapacity
    Output "Enter Initial Passenger Count:"
    Input passengerCount
    // Main Passenger Boarding and Capacity Enforcement Loop
    While boardMore == "YES" AND passengerCount < maxCapacity Do
        Output "Enter Distance Zone (1: Short, 2: Medium, 3: Long):"
        Input distanceZone
        // Input Validation Loop for Distance Zone
        While distanceZone < 1 OR distanceZone > 3 Do
            Output "INVALID ZONE! Enter 1, 2, or 3:"
            Input distanceZone
        End While
        Output "Is this Peak Rush Hour? (YES/NO):"
        Input peakHour

        // Base Fare Determination Based on Tiered Zones
        If distanceZone == 1 Then
            baseFare = 5.00
        Else If distanceZone == 2 Then
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        baseFare = 10.00
    Else
        baseFare = 15.00
    End If
    // Surcharge Calculation for Peak Hours
    If peakHour == "YES" Then
        finalFare = baseFare * 1.25
    Else
        finalFare = baseFare
    End If
    // Increment Passenger State
    passengerCount = passengerCount + 1
    // Boarding Summary Output
    Output "BOARDING SUCCESSFUL!"
    Output "Route: ", routeID
    Output "Fare Due: Le ", finalFare
    Output "Current Passengers: ", passengerCount, " / ", maxCapacity
    // Capacity Threshold Evaluation
    If passengerCount >= maxCapacity Then
        boardMore = "NO"
        Output "ALERT: Vehicle is FULL. Boarding stopped."
    Else
        Output "Board another passenger? (YES/NO):"
        Input boardMore
    End If
  End While
  Output "Session finished. Have a safe journey!"
End CommuterTransitSystem.
```